

A boundary-field gap for the spin-one Heisenberg chain

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Abstract

We prove a uniform spectral gap for odd open spin-one antiferromagnetic Heisenberg chains with the same fixed magnetic field at both endpoints. The field $h = 3/5$ gives a gap greater than $\log(10)/392$ on every chain of $2L + 1$ sites with $L \geq 960$. Tasaki's index theorem then gives index -1 for every subsequential local limit of these boundary-selected ground states.

1 Introduction

Equal positive magnetic fields at the endpoints of an odd open spin-one Heisenberg chain select a unique finite-volume ground state [20, Lemma 1]. The remaining question is whether the excitation gap stays positive as the chain grows. We prove a quantitative gap for one fixed field in the pure bilinear model. This is the boundary formulation used by Tasaki to connect the Haldane gap problem to a nontrivial topological index [20].

In the orthonormal basis e_m , $m = -1, 0, 1$, of $\mathbb{C}^3$, let

$$S^z e_m = m e_m, \quad S^+ e_m = \sqrt{2 - m(m+1)} e_{m+1} \quad (m < 1), \quad S^+ e_1 = 0,$$

and put $S^- = (S^+)^*$, $S^x = (S^+ + S^-)/2$, and $S^y = (S^+ - S^-)/(2i)$. Thus $\sum_\alpha (S^\alpha)^2 = 2I$. For $L \geq 1$ and $h > 0$, consider the operator on $(\mathbb{C}^3)^{\otimes(2L+1)}$ given by

$$H_L(h) = \sum_{j=-L}^{L-1} \sum_{\alpha=x,y,z} S_j^\alpha S_{j+1}^\alpha - h(S_{-L}^z + S_L^z). \quad (1.1)$$

There is no bond between the endpoints. Write $E_0(L, h) \leq E_1(L, h)$ for its two lowest eigenvalues, counted with multiplicity. We address the fixed-boundary-field gap assertion: for some single $h > 0$, the difference $E_1(L, h) - E_0(L, h)$ stays bounded away from zero as $L \rightarrow \infty$.

Theorem 1.1. *For the Hamiltonian (1.1), with the spin normalization above,*

$$E_1(L, 3/5) - E_0(L, 3/5) > \frac{\log 10}{392} \quad \text{for every integer } L \geq 960.$$

Context and prior work. Haldane's prediction, formulated in 1981 and published in 1983, distinguishes integer and half-odd-integer isotropic antiferromagnetic Heisenberg chains, with a positive bulk gap for positive integer spin [8, 9, 10]. For spin one-half, Lieb, Schultz and Mattis had already constructed excitations of the periodic chain whose energy above the ground state vanishes with its length [13, Appendix B]. The Affleck–Kennedy–Lieb–Tasaki construction gave a rigorous gapped spin-one model [1, 2]. Its interaction contains a biquadratic term: with $q = \mathbf{S}_j \cdot \mathbf{S}_{j+1}$, it

is proportional to $q + q^2/3$ up to an additive constant. Our Hamiltonian has the pure bilinear interaction q and fixed endpoint fields.

The endpoint condition matters even when the bulk interaction is fixed. White and Huse numerically characterized the effective spin-one-half degrees of freedom at free ends [21]. To estimate bulk properties, including the gap, they added physical spin-one-half endpoint sites and adjusted their couplings. Magnetic fields on the original spin-one endpoints give a different finite-volume model, whose gap requires its own boundary estimates.

Tasaki formulated precisely this boundary problem in his work on the topological character of the spin-one Heisenberg ground state [20]. Equation (2) of that paper is the Hamiltonian (1.1); Lemma 1 proves uniqueness for every $L \geq 1$ and $h > 0$. Assumption 2 asks for a positive gap uniform in all sufficiently large L , at one fixed $h > 0$. Theorem 1.1 resolves this fixed-boundary-field Haldane gap assertion positively and establishes that assumption. Section 7 then derives Tasaki index -1 for every boundary-selected subsequential local limit.

The topological viewpoint developed through several distinct descriptions. Den Nijs and Rommelse introduced nonlocal string order [4], and Kennedy and Tasaki connected it to hidden symmetry breaking [11]. Pollmann, Turner, Berg and Oshikawa described symmetry protection through matrix-product states and the entanglement spectrum [18]; Ogata extended the bond-centered reflection index to infinite-chain states beyond the matrix-product description [15]. Tasaki’s local-twist index [19] instead uses the twist expectation studied by Nakamura and Todo [14]. The boundary-state theorem in [20] uses uniform z rotations and site-centered reflection. These invariances imply invariance under site reflection combined with a global π rotation about z , the symmetry studied by Fuji, Pollmann and Oshikawa [7]. Our consequence concerns Tasaki’s index in this precise symmetry setting; it does not identify the different diagnostics above.

Methodological inputs. For frustration-free chains, Lemm and Mozgunov proved finite-size criteria involving both interior and boundary-segment gaps [12]. Here the boundary information comes from weighted spatial moments and a coupled Gibbs-purity recurrence. The finite trial vectors use the matrix-product framework of finitely correlated states [5] and its finite-chain formulation [17]; they certify energy upper bounds that initialize the recurrence.

The continuous-history representation is an instance of the time-ordered Poisson expansion of Aizenman and Nachtergaele [3]. The particular spatial transfer operator and finite periodic estimates used here come from [16], which proves a uniform gap on even periodic rings. We retain that source’s spin normalization, coupling, and energy shift, and state explicitly the transfer and finite-input results used below. The endpoint pairings and all estimates specific to the open chain are proved here.

Proof strategy. For each inverse temperature $b > 0$, a partition function of the periodic model is a trace of a power of a compact self-adjoint spatial operator X_b . For the open chain with *opposite* endpoint fields, the corresponding partition function is instead a boundary pairing $\langle v_b, X_b^n v_b \rangle$, where n is the number of sites. At even n , this is a positive weighted moment of the absolute spatial eigenvalues. A rotation of the right boundary vector produces the same-field Hamiltonian in (1.1).

The periodic estimates show that one spatial eigenvalue dominates sufficiently high even moments. For the open chain, however, the contribution of each eigenspace is weighted by its overlap with v_b , and this vector changes with temperature. Periodic concentration therefore leaves a boundary-overlap problem. We solve it by contracting the residual boundary-weighted moments while tracking the physical Gibbs purity (the sum of the squared Gibbs probabilities)

of the opposite-field chain. These two estimates improve together when both length and inverse temperature double, without assuming a nonzero boundary overlap at the next temperature. Once the leading spatial line dominates the boundary pairing, its rotation invariance controls the same-field expression at every odd length in a full dyadic interval. The purity defect then decreases at a rate exponential in the inverse temperature, which yields a uniform physical gap.

Sections 2 and 3 establish the transfer representation and the periodic estimates in the form needed here. Section 4 proves the boundary recurrence. Section 5 initializes and iterates it, and Section 6 proves Theorem 1.1 at all the required odd lengths. Section 7 applies the resulting gap to the boundary-selected infinite-volume states. Appendices A and B give complete finite certificates for the boundary initialization. All terminating decimals used in inequalities denote exact rationals.

2 Spatial transfer with endpoint fields

We first put the periodic and open chains in the same transfer representation. The boundary vectors, including their complex phases, will distinguish the physical endpoint fields. The time-ordered bond expansion has a general formulation in [3, arXiv v1, Section 2.2, Equations (2.10)–(2.12)]. We prove the required boundary identity directly with the specified kernel and field coefficients.

2.1 Hamiltonians and shifted partition functions

From now on set

$$h = \frac{3}{5}, \quad a = \frac{700741}{500000}, \quad b_0 = \frac{49}{4}, \quad b_j = 2^j b_0 \quad (j \geq 0). \quad (2.1)$$

It is convenient to label an n -site chain by $0, \dots, n-1$. Write $S_j \cdot S_k = \sum_{\alpha} S_j^{\alpha} S_k^{\alpha}$ and define

$$H_n^{\text{opp}} = \sum_{j=0}^{n-2} S_j \cdot S_{j+1} - h S_0^z + h S_{n-1}^z, \quad (2.2)$$

$$H_n^{\text{s}} = \sum_{j=0}^{n-2} S_j \cdot S_{j+1} - h S_0^z - h S_{n-1}^z, \quad (2.3)$$

$$H_n^{\text{per}} = \sum_{j=0}^{n-2} S_j \cdot S_{j+1} + S_{n-1} \cdot S_0 \quad (n \geq 4). \quad (2.4)$$

For $n = 2L + 1$, H_n^{s} is (1.1) after relabeling. Our partition functions are ordinary traces over the entire physical Hilbert space:

$$\begin{aligned} Z_n(b) &= \text{Tr} e^{-b(H_n^{\text{per}} + anI)}, \\ Y_n(b) &= \text{Tr} e^{-b(H_n^{\text{opp}} + anI)}, \quad Y_n^{\text{s}}(b) = \text{Tr} e^{-b(H_n^{\text{s}} + anI)}. \end{aligned} \quad (2.5)$$

The shift is per site, also for open chains. It will cancel from all physical Gibbs weight ratios.

2.2 The spatial operator

We use the concrete spatial-transfer operator in [16, Proposition 3.1]. Its auxiliary space records ordered bond events in imaginary time; each physical site contributes a kernel depending on its

two incident histories. Use the unitarily equivalent Cartesian spin matrices $(S^\alpha)_{rs} = -i\varepsilon_{\alpha rs}$ and set $G_\alpha = iS^\alpha$. Then G_α is real, $\|G_\alpha\| = 1$, and

$$-S \cdot S = \sum_{\alpha=x,y,z} G_\alpha \otimes G_\alpha.$$

A history is an ordered list $\omega = ((t_1, \alpha_1), \dots, (t_d, \alpha_d))$ with $0 < t_1 < \dots < t_d < b$ and $\alpha_r \in \{x, y, z\}$; the empty list is allowed. Give the disjoint union Ω_b of these labeled simplexes Lebesgue measure on each simplex and mass one at the empty history. Its measure μ_b has total mass e^{3b} .

For two histories, merge their events in chronological order and take the trace of the resulting product of G matrices, with increasing time from left to right. Denote this real kernel by $k_b(\omega, \eta)$. Set it to zero at nonempty time collisions, a null set, and use trace 3 for the two empty histories. The kernel is symmetric: exchanging the two histories leaves their merged list unchanged. In particular, symmetry does not reverse a product of noncommuting matrices. Moreover $|k_b| \leq 3$. On $\mathcal{K}_b = L^2(\Omega_b, \mu_b; \mathbb{C})$ define

$$(X_b f)(\omega) = e^{-ab} \int_{\Omega_b} k_b(\omega, \eta) f(\eta) d\mu_b(\eta). \quad (2.6)$$

It is real, self-adjoint, and Hilbert–Schmidt. The cited transfer Proposition identifies its periodic moments:

$$Z_n(b) = \text{Tr } X_b^n \quad (n \geq 4), \quad \sum_i |\mu_i(b)|^k < \infty \quad (k \geq 2), \quad (2.7)$$

where $\mu_i(b)$ lists its nonzero eigenvalues with multiplicity. This exact operator, with the shift in (2.1), is the one used for the periodic inputs in Section 3.

Rotations will let us change one endpoint field and, later, show that the leading spatial line is fixed. Let D_2 consist of the identity and the three diagonal rotations by π , let C cyclically permute the coordinate axes, and put $\mathcal{G} = \langle D_2, C \rangle$. This is a group of twelve proper signed permutations. If $ge_\alpha = \epsilon_g(\alpha)e_{\sigma_g(\alpha)}$, write $\sigma_g\omega$ for relabeling the marks and $s_g(\omega) = \prod_r \epsilon_g(\alpha_r)$. The operators

$$(U_g f)(\omega) = s_g(\sigma_g^{-1}\omega) f(\sigma_g^{-1}\omega) \quad (2.8)$$

form a real unitary representation and commute with X_b . This follows either directly by rotating every matrix in the kernel, or from the same Proposition. In particular, fix $P = \text{diag}(1, -1, -1)$, so $Pe_z = -e_z$.

2.3 Boundary vectors

Define $v_b \in \mathcal{K}_b$ by

$$v_b(\omega) = \begin{cases} (ih)^d, & \text{all } d \text{ marks of } \omega \text{ equal } z, \\ 0, & \text{otherwise.} \end{cases} \quad (2.9)$$

The empty-history value is one. The simplex volumes give $\|v_b\|^2 = \sum_{d \geq 0} b^d h^{2d}/d! = e^{bh^2}$. The inner product below is conjugate linear in its first argument.

Proposition 2.1 (Open-chain transfer). *For $b > 0$, $n \geq 2$, and $g \in \mathcal{G}$, the pairing $\langle v_b, X_b^n U_g v_b \rangle$ is the shifted partition function of the open chain with left field $-hS_0^z$ and right field $+hS_{n-1}^{ge_z}$. Consequently*

$$Y_n(b) = \langle v_b, X_b^n v_b \rangle, \quad Y_n^s(b) = \langle v_b, X_b^n U_P v_b \rangle. \quad (2.10)$$

Here $S^u = \sum_\alpha u_\alpha S^\alpha$ for a real vector u .

Proof. Expand the negative Hamiltonian exponential into its ordered-time series. Each bond event inserts two G_α matrices, by the Cartesian identity above. At the left endpoint, the conjugated coefficient in (2.9) gives $(-ih)G_z = +hS^z$ per event; at the right endpoint the coefficient is $(ih)G_z = -hS^z$. Relabeling by g and multiplying its signs changes this last insertion to $-hS^{ge_z}$. These are exactly the two field insertions in the negative Hamiltonian stated in the Proposition.

Regroup the series by the boundary histories ω_0, ω_n and the $n-1$ bond histories $\omega_1, \dots, \omega_{n-1}$. Their interleavings partition each full time simplex up to null collisions. At site r , $0 \leq r < n$, the trace is $k_b(\omega_r, \omega_{r+1})$, in the original chronological order. Integrating these n kernel factors against the two boundary coefficients gives X_b^n and the shift factor e^{-ban} . No bond joins the endpoints.

This regrouping is legitimate: the absolute sum of traced terms is bounded by $3^n \exp\{b[3(n-1) + 2h]\}$ before the scalar shift. All sums and integrals are therefore absolutely convergent. Choosing $g = 1$ and $g = P$ gives (2.10). $\square$

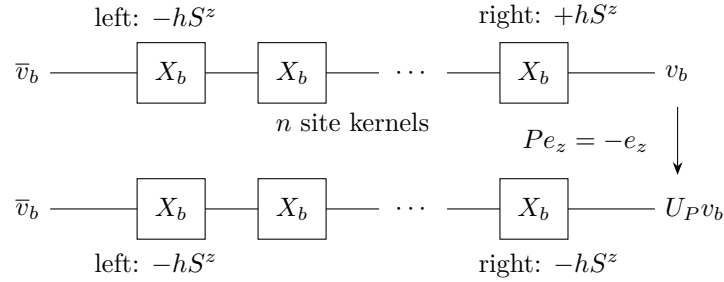


Figure 1: The two spatial boundary pairings. Each box contributes one site kernel; the internal integrations represent the $n-1$ bonds. Replacing the right boundary vector by $U_P v_b$ changes the right physical field. This is an identity between transfer representations, not a claim that the two physical Hamiltonians are unitarily equivalent.

For even lengths, the opposite-field expression is the positive moment $Y_n(b) = \langle v_b, |X_b|^n v_b \rangle$. At odd lengths, the powers of X_b retain the signs of its eigenvalues. We first concentrate the even opposite-field moments. The resulting leading-line estimate will then control the signed same-field pairing at odd lengths; Figure 1 distinguishes the two boundary pairings.

3 Periodic concentration estimates

The boundary argument requires quantitative information about the spatial spectrum, including the change of its leading eigenvalue when the inverse temperature doubles. We extract that information from finite periodic-chain estimates in [16]. The extraction is given in detail because the periodic gap theorem alone does not provide these boundary estimates. Throughout, terminating decimals denote exact rational numbers, and eigenvalue lists count multiplicities.

For even $n \geq 4$, put

$$S(n, b) = \frac{Z_{2n}(b)}{Z_n(b)^2}, \quad T(n, b) = \frac{Z_n(2b)}{Z_n(b)^2}. \quad (3.1)$$

The first is the purity of the spatial probabilities $|\mu_i(b)|^n / Z_n(b)$, where $(\mu_i(b))$ is the nonzero spectrum of X_b ; the second is the purity of the physical Gibbs probabilities. Both lie in $(0, 1]$. Doubling n squares the first probability list, whereas doubling b squares the second.

Lemma 3.1 (Squaring and concentration). *Let (w_i) be a finite or countable probability list, and suppose its purity is at least $1 - u$, where $0 \leq u < 1$. Normalizing the squares w_i^2 gives a list with purity defect at most*

$$f(u) = \frac{u^2}{2(1-u)^2}. \quad (3.2)$$

If $u < 1/2$, the largest weight is unique and at least $1 - u$. Writing x for the sum of all the other weights, one has

$$x \leq \frac{1 - \sqrt{1 - 2u}}{2}. \quad (3.3)$$

In particular, for $0 \leq u \leq .1$,

$$\frac{x}{1-x} \leq d(u), \quad d(u) = \frac{u}{2-3u}. \quad (3.4)$$

Proof. Set $P = \sum_i w_i^2$ and let P' be the purity after squaring. Nonnegativity justifies rearrangement of the countable sums, and

$$1 - P' = \frac{2 \sum_{i < k} w_i^2 w_k^2}{P^2} \leq \frac{2(\sum_{i < k} w_i w_k)^2}{P^2} = \frac{(1 - P)^2}{2P^2} \leq f(u).$$

Here f is increasing on $[0, 1)$. A summable nonzero nonnegative list attains its maximum, and $P \leq \max_i w_i$. If $u < 1/2$, the maximum therefore exceeds $1/2$ and is unique. Thus $x \leq u < 1/2$, and

$$2x(1-x) \leq 1 - P \leq u,$$

which gives (3.3). Also $x \leq u/[2(1-x)] \leq u/[2(1-u)]$; applying the increasing function $x \mapsto x/(1-x)$ proves (3.4). $\square$

For later reference we also record explicitly the interaction of the two periodic purities. This is [16, Lemma 4.2]; we include its short proof by partition-function cancellation.

Lemma 3.2 (Coupled update). *Suppose that $n \geq 4$ is even, $b > 0$, and $S(n, b) \geq s$, $T(2n, b) \geq t$, with $0 < s, t \leq 1$. Set*

$$p_+ = f(1 - s^2 t).$$

If $p_+ < 1$, then

$$1 - S(2n, 2b) \leq p_+, \quad 1 - T(4n, 2b) \leq f(1 - t^2(1 - p_+)). \quad (3.5)$$

Either upper defect bound may be increased before subsequent use, provided it stays below one.

Proof. Directly from (3.1),

$$S(n, 2b) = S(n, b)^2 \frac{T(2n, b)}{T(n, b)^2} \geq s^2 t.$$

Apply spatial squaring and Lemma 3.1. A second cancellation then gives

$$T(4n, b) = T(2n, b)^2 \frac{S(2n, 2b)}{S(2n, b)^2} \geq t^2(1 - p_+).$$

Physical squaring proves the second inequality. Monotonicity of f justifies increasing either defect bound. $\square$

We next state the finite inputs gathered in the companion [16, Corollary 5.6]. Its shift, spin normalization and unit bilinear coupling are exactly those used here. In particular, we use an intermediate row in the initialization proof, not merely its final conclusion.

Proposition 3.3 (Finite periodic inputs from [16]). *Set*

$$\ell = .999, \quad U = 1.00139, \quad V = .872, \quad m_J = 1.0657205, \quad m_N = .2783155. \quad (3.6)$$

At $b_0 = 49/4$, the spatial eigenvalue list decomposes as one list J and three identical lists N . Here J is the spectrum on the subspace fixed by every U_g , $g \in D_2$, and N is the common spectrum on each of the other three simultaneous eigenspaces of these commuting operators. In J the absolute eigenvalues are at most U , and their twelfth powers sum to at most m_J ; in each N they are at most V , and their twelfth powers sum to at most m_N . There is an eigenvalue in J whose modulus exceeds ℓ . Furthermore,

$$1 - S(72, b_1) \leq \frac{151249}{2500000} = .0604996. \quad (3.7)$$

For the periodic chains of 72 and 120 sites, the shifted ground energy is strictly negative. Consequently, at each of these lengths and every $b > 0$, an individual unnormalized ground Boltzmann weight is greater than one.

No boundary Hamiltonian occurs in Proposition 3.3. The finite periodic thermal and variational certificates supporting it are given in the cited source. We now deduce all the spatial estimates that will be used for boundaries. The residual ratios will contract the boundary-weighted tails. The upper and lower bounds comparing successive leading eigenvalues will control the change of temperature in the boundary concentration and physical-purity estimates, respectively. Finally, positivity and rotation invariance of the leading line will control the same-field pairing at odd lengths.

Proposition 3.4 (Concentration at the doubling scales). *For every $j \geq 0$, X_{b_j} has a unique simple eigenvalue of largest modulus. This eigenvalue is positive; denote it by $\lambda_j = \|X_{b_j}\|$. Its eigenspace is fixed pointwise by $\mathcal{G}$. Let ρ_j be the supremum of the absolute values of all other eigenvalues divided by λ_j . Then*

$$\ell < \lambda_0 \leq U, \quad \rho_0 < .874, \quad \rho_1 < .9525, \quad \rho_2 < .971. \quad (3.8)$$

For $j \geq 2$, define

$$n_j = 120 \cdot 2^{j-2}, \quad M_j = 2n_j, \quad u_2 = .052, \quad u_{j+1} = 6u_j^2. \quad (3.9)$$

The following bounds hold for every $j \geq 2$:

$$1 - S(n_j, b_j) \leq u_j, \quad 1 - T(M_j, b_j) \leq u_j, \quad (3.10)$$

$$\rho_j^{n_j} \leq d(u_j). \quad (3.11)$$

Finally, put $r_j = \lambda_{j+1}/\lambda_j^2$. Then

$$0 < r_j \leq 1 \quad (j \geq 0), \quad r_j^{M_j} \geq (1 - u_j)(1 - u_{j+1}) \quad (j \geq 2). \quad (3.12)$$

Proof. Until positivity has been proved, write $\lambda_j = \|X_{b_j}\|$ for the maximal spectral modulus.

The initial spatial spectrum. Remove an eigenvalue in J whose modulus exceeds ℓ . The remaining twelfth-power mass in J is at most $m_J - \ell^{12} < \ell^{12}$, while $V < \ell$ bounds each of the other three sectors. The removed eigenvalue is therefore the unique simple eigenvalue of maximal modulus. The exact rational inequalities

$$m_J - \ell^{12} < (.874\ell)^{12}, \quad V < .874\ell$$

prove $\rho_0 < .874$.

We need the total residual moments as well as this individual ratio bound. For every real $k \geq 12$, define

$$D(k) = \frac{(m_J - \ell^{12})^{k/12} + 3[V^k + (m_N - V^{12})^{k/12}]}{\ell^k}. \quad (3.13)$$

The sum of the k th powers of all nondominant absolute eigenvalue ratios at b_0 is at most $D(k)$. Indeed, $V^{12} < m_N < 2V^{12}$; for a nonnegative list with entries at most V^{12} and total mass at most m_N , convexity of $x^{k/12}$ bounds the sum by $(V^{12})^{k/12} + (m_N - V^{12})^{k/12}$. This follows by moving mass to fill one entry before the next; finite sublists followed by a limit give the countable case. Without an entry cap, the corresponding bound is simply the total mass to the power $k/12$. Applying these facts to the four sectors proves the claim. In particular, for even $k \geq 12$,

$$Z_k(b_0) \leq U^k(1 + D(k)). \quad (3.14)$$

Reseeding at $(120, b_2)$. Our next goal is to make both $1 - S(120, b_2)$ and $1 - T(240, b_2)$ at most .052, so that one quadratic recurrence will control them. A coupled update will give these bounds from spatial and physical seeds at $(60, b_1)$ and $(120, b_1)$, respectively. To obtain the spatial seed, we combine a leading-eigenvalue lower bound from length 72 with a residual-moment upper bound at length 32, then increase the moment order to 60. The same residual estimate also supplies the ratio ρ_1 needed in the first boundary updates.

By (3.7) and Lemma 3.1, the spatial list at $(72, b_1)$ has a unique largest weight. Thus X_{b_1} also has a unique simple eigenvalue of maximal modulus. Moreover,

$$.0604996 < 2(.032)(.968)$$

implies that this weight exceeds .968. The periodic trial at length 72 gives $Z_{72}(b_1) > 1$, so

$$\lambda_1^{72} > .968. \quad (3.15)$$

Physical squaring at length 32 gives $Z_{32}(b_1) \leq Z_{32}(b_0)^2$. Subtracting the dominant spatial contribution and using (3.14) and (3.15) therefore yields

$$\sum_{i \neq 0} \left| \frac{\mu_i(b_1)}{\lambda_1} \right|^{32} \leq U^{64}(1 + D(32))^2(.968)^{-32/72} - 1 < .9525^{32}. \quad (3.16)$$

This proves $\rho_1 < .9525$. For any nonnegative list (a_i) and $q \geq p > 0$, one has $\sum_i a_i^q \leq (\sum_i a_i^p)^{q/p}$ whenever the right side is finite. Applying this to (3.16) gives a residual 60th moment below $.9525^{60}$. Retaining only the dominant term in the numerator of $S(60, b_1)$ proves

$$S(60, b_1) \geq s := (1 + .9525^{60})^{-2}. \quad (3.17)$$

For the physical spectrum at length 120, choose one ground Boltzmann weight at b_0 and divide every other weight by it. The chosen weight is greater than one by Proposition 3.3. Hence the sum of the other ratios is bounded above by

$$R = U^{120}(1 + D(120)) - 1.$$

When b doubles, these ratios square, so their sum is at most R^2 . Keeping the chosen ground weight in the purity numerator now gives

$$T(120, b_1) \geq t := (1 + R^2)^{-2}. \quad (3.18)$$

This reasoning neither assumes ground-state simplicity nor discards multiplicities.

Here are rational checks sufficient for every numerical comparison just used and for the ensuing seed:

$$\begin{aligned} D(32) &< .0442, \\ D(120) &< .00000025, \\ (.9856)^9 &< (.968)^4, \\ U^{64}(1 + .0442)^2 / .9856 - 1 &< .2092 < .9525^{32}, \\ s &> .9002, \quad t > .9372, \\ f(1 - .9002^2 .9372) &< .0502, \\ f(1 - .9372^2(1 - .0502)) &< .0198. \end{aligned} \quad (3.19)$$

All comparisons are between explicit rational numbers after bounding the cube roots in $D(32)$ by

$$(m_J - \ell^{12})^{1/3} < .426635, \quad (m_N - V^{12})^{1/3} < .439736.$$

Cubing verifies these two bounds; raising the bounds to the eighth power gives the displayed bound on $D(32)$. For $D(120)$ no roots are needed. The bound on t follows already with $D(120)$ replaced by .00000025. Lemma 3.2, applied to (3.17)–(3.18) at $(n, b) = (60, b_1)$, now gives

$$1 - S(120, b_2) < .052, \quad 1 - T(240, b_2) < .052.$$

Induction. For $0 \leq u \leq .052$,

$$f(1 - (1 - u)^3) = G(u)u^2, \quad G(u) = \frac{1}{2} \left(\sum_{r=1}^3 (1 - u)^{-r} \right)^2. \quad (3.20)$$

The function G is increasing, and exact substitution gives $G(.052) < 5.583 < 6$. Also $6u^2 \leq u$ on this interval. Thus, if the two defects in (3.10) are at most u_j , the first coupled output is at most $6u_j^2 = u_{j+1} \leq u_j$. Using this upper bound in the second output bounds that output by the same number. This proves (3.10) for every $j \geq 2$. Each spatial purity exceeds $1/2$, so the maximal spatial modulus remains unique and simple. Applied to its probability list, Lemma 3.1 also gives

$$\sum_{i \neq 0} \left| \frac{\mu_i(b_j)}{\lambda_j} \right|^{n_j} \leq d(u_j),$$

which proves (3.11). The rational comparison $d(.052) < .971^{120}$ gives $\rho_2 < .971$.

The sign and the rotation symmetry. At every level we have proved uniqueness of the eigenvalue of maximal modulus. If it were $-\lambda_j$, then $\text{Tr}(X_{b_j}^n)/\lambda_j^n$ would tend to -1 through odd integers n . Indeed, the residual ratios are bounded by some number below one, and their fourth powers

have finite sum, so their n th-power sum vanishes as $n \rightarrow \infty$. This contradicts the strictly positive physical partition function $Z_n(b_j)$ at odd $n \geq 5$. The leading eigenvalue is therefore $+\lambda_j$.

Its eigenspace has a real unit vector, since X_{b_j} is real. The real unitaries U_g commute with X_{b_j} , so they act on that line by signs. The order-three cyclic rotation C must act by $+1$. It conjugates the three nonidentity elements of D_2 cyclically, so they have the same sign on the line. Since the product of any two is the third, that common sign equals its square and hence is $+1$. Thus every element of $\mathcal{G} = \langle D_2, C \rangle$ fixes the leading eigenspace pointwise.

Comparing the leading eigenvalues. For every even $N \geq 4$, physical squaring gives $Z_N(b_{j+1}) \leq Z_N(b_j)^2$. The residual-tail estimate used above gives $Z_N(b_j)/\lambda_j^N = 1 + o(1)$, so $Z_N(b_j)^{1/N} \rightarrow \lambda_j$. Taking N th roots and letting N tend to infinity through even integers therefore yields $\lambda_{j+1} \leq \lambda_j^2$. For the opposite direction at the reference scale, recall that $M_j = n_{j+1}$. The leading spatial probability at (M_j, b_{j+1}) is at least $1 - u_{j+1}$, so

$$\begin{aligned} \lambda_{j+1}^{M_j} &\geq (1 - u_{j+1})Z_{M_j}(b_{j+1}) \\ &= (1 - u_{j+1})T(M_j, b_j)Z_{M_j}(b_j)^2 \\ &\geq (1 - u_{j+1})(1 - u_j)\lambda_j^{2M_j}. \end{aligned}$$

This proves (3.12) and completes the proof. $\square$

4 Concentration of boundary-weighted moments

We now pass from unweighted spatial traces to the boundary pairing for the opposite-field chain. We will control two quantities together: the fraction of an even boundary moment outside the leading spatial line, and the defect of the physical Gibbs purity. The boundary coefficients depend on temperature. We compare the weighted moments across temperatures through physical partition-function identities, while contracting the residual spatial powers at a fixed temperature.

Use the simple positive leading eigenvalue λ_j and residual ratio ρ_j from Section 3. Let P_j be the orthogonal projection onto its line, and put

$$\begin{aligned} c_j &= \|P_j v_{b_j}\|^2, \\ B_k(j) &= \langle v_{b_j}, (|X_{b_j}|/\lambda_j)^k v_{b_j} \rangle, \quad e_k(j) = \frac{B_k(j) - c_j}{B_k(j)}, \\ A(m, j) &= \frac{Y_m(b_{j+1})}{Y_m(b_j)^2}. \end{aligned} \tag{4.1}$$

We use k, m even and at least two. Then $B_k(j) = Y_k(b_j)/\lambda_j^k > 0$, so $e_k(j)$ is defined even if $c_j = 0$. Also $0 \leq e_k(j) \leq 1$ and $0 < A(m, j) \leq 1$. Thus $e_k(j)$ is the residual fraction of a spatial boundary moment, whereas $A(m, j)$ is the physical Gibbs purity of the opposite-field chain at inverse temperature b_j .

The normalized absolute spatial eigenvalues are at most one. Thus $B_m(j) \leq B_k(j)$ when $m \geq k$. More precisely, if $D_k(j) = B_k(j) - c_j$, the spectral theorem gives

$$0 \leq D_m(j) \leq \rho_j^{m-k} D_k(j) \quad (m \geq k). \tag{4.2}$$

The powers act on the residual spectrum; no upper bound for the total boundary norm is needed in this comparison.

Lemma 4.1 (Transfer between temperatures). *Let $j \geq 0$ and let $m > k \geq 2$ be even. Suppose $e_k(j) < 1$ and $0 < q < 1$ with $q \geq \rho_{j+1}^{m-k}$. Define, for $0 \leq e < 1$ and $0 < t \leq 1$,*

$$F(q, e, t) = \frac{q}{1-q} \left(\frac{1}{(1-e)^2 t} - 1 \right). \quad (4.3)$$

Then

$$e_m(j+1) \leq F(q, e_k(j), A(m, j)). \quad (4.4)$$

In particular, an upper bound strictly below one on the right proves $c_{j+1} > 0$ without presupposing it.

Proof. At the new temperature, (4.2) yields $B_m(j+1) - c_{j+1} \leq q[B_k(j+1) - c_{j+1}]$. Subtract and rearrange to obtain

$$e_m(j+1) \leq \frac{q}{1-q} \left(\frac{B_k(j+1)}{B_m(j+1)} - 1 \right).$$

Write $r_j = \lambda_{j+1}/\lambda_j^2$. From the definitions,

$$\frac{B_k(j+1)}{B_m(j+1)} = \frac{A(k, j)}{A(m, j)} \left(\frac{B_k(j)}{B_m(j)} \right)^2 r_j^{m-k}. \quad (4.5)$$

Section 3 gives $r_j \leq 1$, and physical purity gives $A(k, j) \leq 1$. The hypothesis $e_k(j) < 1$ implies $c_j > 0$, and $B_m(j) \geq c_j$ gives $B_k(j)/B_m(j) \leq (1 - e_k(j))^{-1}$. Substitution proves (4.4). Only the old overlap c_j has been divided by. The new overlap follows from $e_m(j+1) < 1$ and $B_m(j+1) > 0$. $\square$

To couple this residual estimate to physical purity, we use a shorter reference length for the boundary defect. Recall $M_j = 240 \cdot 2^{j-2}$ and u_j from Proposition 3.4. For $j \geq 2$ let

$$k_2 = 142, \quad k_{j+1} = M_j. \quad (4.6)$$

We track upper bounds E_j on $e_{k_j}(j)$ and η_j on $1 - A(M_j, j)$. The difference $M_j - k_j$ supplies residual contraction at the next temperature. Its output at length $k_{j+1} = M_j$ then controls the ratio $B_{2M_j}(j+1)/B_{M_j}(j+1)$ needed to improve physical purity. The notation η_j denotes only this physical purity defect.

Proposition 4.2 (Coupled boundary update). *Suppose $j \geq 2$, $0 \leq E_j, \eta_j < 1$, and*

$$e_{k_j}(j) \leq E_j, \quad A(M_j, j) \geq 1 - \eta_j.$$

Let $E_{j+1}, \eta_{j+1} \in [0, 1)$ satisfy

$$E_{j+1} \geq F(d(u_{j+1})^{(M_j-k_j)/M_j}, E_j, 1 - \eta_j), \quad (4.7)$$

$$\eta_{j+1} \geq f(1 - (1 - \eta_j)(1 - u_j)(1 - u_{j+1})(1 - E_{j+1})). \quad (4.8)$$

Then $e_{k_{j+1}}(j+1) \leq E_{j+1}$ and $A(M_{j+1}, j+1) \geq 1 - \eta_{j+1}$.

Proof. Here $M_j > k_j$ and $n_{j+1} = M_j$. The periodic residual bound gives

$$\rho_{j+1}^{M_j-k_j} \leq d(u_{j+1})^{(M_j-k_j)/M_j} < 1.$$

The function F increases in q, e and decreases in t on the stated domains. Lemma 4.1 therefore proves (4.7)'s conclusion, including $c_{j+1} > 0$.

Put $M = M_j$. Exact cancellation in (4.1) gives

$$\frac{A(2M, j)}{A(M, j)} = r_j^M \frac{B_{2M}(j+1)}{B_M(j+1)} \left(\frac{B_M(j)}{B_{2M}(j)} \right)^2. \quad (4.9)$$

The squared factor is at least one, and the first boundary ratio is at least $c_{j+1}/B_M(j+1) \geq 1 - E_{j+1}$. Thus Proposition 3.4 gives

$$A(2M, j) \geq (1 - \eta_j)(1 - u_j)(1 - u_{j+1})(1 - E_{j+1}).$$

At the fixed physical length $2M$, passing from b_j to b_{j+1} normalizes the squares of the Gibbs probabilities. Lemma 3.1 bounds the resulting defect by f of the preceding defect. Since $M_{j+1} = 2M$, this is (4.8). $\square$

5 Finite initialization and quadratic improvement

We now initialize the boundary recurrence and show that it continues indefinitely. The sixth boundary moment starts a sequence of residual estimates at $(34, b_0)$, $(78, b_1)$, and $(142, b_2)$. A physical-purity estimate at $(240, b_2)$ then starts the coupled update. The inputs are four variational energies and thermal moments on at most six sites.

Proposition 5.1 (Finite boundary data). *Let $\alpha = .4042$ and $P_*(x) = x^3 - .14x^2 - .18x$. For the opposite-field Hamiltonian (2.2),*

$$E_0(H_m^{\text{opp}}) + am < \alpha \quad (m = 34, 78, 142, 240). \quad (5.1)$$

At inverse temperature b_0 ,

$$Y_6(b_0) < .027, \quad c_0 P_*(\lambda_0)^2 < .003745. \quad (5.2)$$

Proof. Proposition B.1 proves (5.1) with explicit integer trial vectors. Proposition A.1 proves (5.2) by finite thermal recurrences with certified rounding errors. The appendices give all matrix and boundary data and the precise finite inequalities. $\square$

5.1 Starting residual and physical-purity bounds

The thermal estimates bound the residual and leading contributions to $Y_m(b_0)$, while the variational estimate supplies a lower bound for one ground weight. Keep ℓ, U from the periodic seed and define, for each of the four lengths in (5.1),

$$\epsilon(m) = .027 e^{b_0 \alpha} U^m \frac{.874^{m-6}}{\ell^6}, \quad R(m) = e^{.132} U^m + \epsilon(m) - 1. \quad (5.3)$$

Lemma 5.2. *At these four lengths,*

$$e_m(0) \leq \epsilon(m), \quad A(m, j) \geq (1 + R(m)^{2^j})^{-2} \quad (j \geq 0). \quad (5.4)$$

Proof. By (4.2), $\rho_0 \leq .874$, and $\ell < \lambda_0 \leq U$,

$$\lambda_0^m D_m(0) \leq U^m .874^{m-6} \frac{Y_6(b_0)}{\ell^6} < e^{-b_0 \alpha} \epsilon(m).$$

The trial bound supplies a physical ground weight greater than $e^{-b_0\alpha}$ and hence $Y_m(b_0) > e^{-b_0\alpha}$. Dividing gives the first inequality.

To bound the full physical partition function from above, factor $P_*(x) = x(x - 1/2)(x + 9/25)$. On $[\ell, U]$ it is positive and $xP'_*(x) < 4P_*(x)$. For example the latter is the positivity of $4P_*(x) - xP'_*(x) = x(x^2 - .28x - .54)$ on this interval. It follows that $x^m/P_*(x)^2$ is increasing for $m > 8$. Consequently (5.2) gives

$$e^{b_0\alpha} \lambda_0^m c_0 \leq U^m \frac{.003745 e^{b_0\alpha}}{P_*(U)^2} < e^{.132} U^m.$$

The last scalar comparison has left coefficient below 1.133226, whereas $e^{.132} > 1.141108$. Combining the leading and residual estimates yields $e^{b_0\alpha} Y_m(b_0) \leq 1 + R(m)$.

Select one physical ground eigenvector; uniqueness is not needed. Its unnormalized weight g is greater than $e^{-b_0\alpha}$, so the sum of the remaining weights divided by g is at most $R(m)$. In particular $R(m) \geq 0$. At $b_j = 2^j b_0$, the corresponding rest-to-ground ratio is at most $R(m)^{2^j}$, since a sum of nonnegative 2^j th powers is at most the 2^j th power of the sum. The purity is at least the square of the normalized weight of this chosen ground eigenvector. This proves the second inequality. $\square$

In particular $\epsilon(34) < .0928 < 1$, proving $c_0 > 0$. The estimates $\rho_1 \leq .953$ and $\rho_2 \leq .971$ now allow two applications of Lemma 4.1:

$$e_{78}(1) \leq F(.953^{44}, .093, (1 + R(78))^{-2}) < .133, \quad (5.5)$$

$$e_{142}(2) \leq F(.971^{64}, .133, (1 + R(142)^2)^{-2}) < .14. \quad (5.6)$$

The bound at $m = 240, j = 2$ gives $1 - A(240, 2) < .21$. Thus the induction of Proposition 4.2 starts with

$$E_2 = .14, \quad \eta_2 = .21. \quad (5.7)$$

Both new leading overlaps have also been proved positive by the strict defect bounds; no overlap was assumed at the next temperature.

5.2 Explicit finite comparisons

We record numerical margins to make the initialization reproducible. The three lower bounds from Lemma 5.2 used above are

$$\begin{aligned} (1 + R(78))^{-2} &> .6180, \\ (1 + R(142)^2)^{-2} &> .7533, \\ (1 + R(240)^4)^{-2} &> .7923. \end{aligned} \quad (5.8)$$

Using the full expressions in (5.3) gives upper outputs below .132129, .137301, and .207518 in (5.5), (5.6), and $1 - A(240, 2)$ respectively. The lower bounds in (5.8) are rounded for display.

Starting from (5.7), the following rounded bounds satisfy Proposition 4.2:

j	E_j	η_j	upper bound on the q used for step $j \rightarrow j + 1$
2	.14	.21	.142
3	.12	.155	.0282
4	.016	.026	.0028
5	.001	.001	—

(5.9)

The exponents $(M_j - k_j)/M_j$ are $98/240$, $1/2$, $1/2$. The successive upper outputs for E_{j+1} are below $.117754$, $.015328$, $.000170$; those for η_{j+1} are below $.147078$, $.025188$, $.000432$. In computing the latter outputs one uses the displayed rounded E_{j+1} , so each line closes with the next.

For clarity, all the scalar comparisons here and in the periodic initialization have an elementary rational verification. For $0 \leq t \leq 10$, put $T_{80}(t) = \sum_{r=0}^{80} t^r/r!$. Then

$$T_{80}(t) \leq e^t \leq T_{80}(t) + 2t^{81}/81!. \quad (5.10)$$

Indeed the ratio of consecutive omitted terms is at most $10/82 < 1/2$. For negative exponents use reciprocal intervals. Rational positive root comparisons can be checked by raising both sides to their integer powers; for example the first q bound in (5.9) is $d(u_3)^{49} < .142^{120}$. Every denominator is positive. These rules, exact rational input values, and the displayed formulas suffice to verify the finite comparisons without floating-point assumptions. The supplementary scalar certificate records all of them.

5.3 Indefinite continuation

We now enlarge the three defect bounds at $j = 5$ to $.01$ so that a single quadratic recurrence controls all later levels.

Proposition 5.3. *Define $x_5 = .01$ and $x_{j+1} = 10x_j^2$ for $j \geq 5$. Then, for every $j \geq 5$,*

$$u_j \leq x_j, \quad e_{k_j}(j) \leq x_j, \quad 1 - A(M_j, j) \leq x_j. \quad (5.11)$$

In particular

$$x_j = 10^{-1-2^{j-5}}. \quad (5.12)$$

Proof. The table proves the two boundary inequalities at $j = 5$, and $u_5 < .000015 < .01$. Suppose (5.11) holds, and write $x = x_j \leq .01$. Here $k_j = M_j/2$ and $u_{j+1} = 6u_j^2 \leq 6x^2$. The first argument of F in (4.7) satisfies

$$\sqrt{d(u_{j+1})} \leq \sqrt{\frac{6x^2}{2 - 18x^2}} \leq 2x.$$

The first output is therefore bounded by $F(2x, x, 1 - x)$. For $x > 0$, division by x^2 gives

$$\frac{F(2x, x, 1 - x)}{x^2} = \frac{2}{1 - 2x}((1 - x)^{-1} + (1 - x)^{-2} + (1 - x)^{-3}).$$

This expression is increasing on $[0, .01]$ after its continuous extension at zero. Its endpoint value is $42430000/6792093 < 10$. Thus we may take $E_{j+1} = 10x^2$.

For factors in $[0, 1]$, the defect of their product is at most the sum of their defects. Thus the four-factor product in (4.8) has defect at most $x + x + 6x^2 + 10x^2 = 2x + 16x^2$. Moreover

$$\frac{f(2x + 16x^2)}{x^2} = \frac{(2 + 16x)^2}{2(1 - 2x - 16x^2)^2}$$

is increasing on $[0, .01]$ and has endpoint value $3645000/1495729 < 10$. Thus we may also take $\eta_{j+1} = 10x^2$. Finally $u_{j+1} \leq 6x^2 \leq 10x^2$. This proves (5.11) at the next level; the sequence remains in $[0, .01]$. Solving its recurrence gives (5.12). $\square$

6 Same fields, odd lengths, and the physical gap

The estimates obtained so far concern even moments of the opposite-field chain. We now use the rotation-invariant leading line to control the same-field partition function at every odd length in a dyadic interval.

Lemma 6.1 (Signed boundary comparison). *For $j \geq 0$, an even $M \geq 2$, and any integer $n \geq M$, define $Q_j(n) = Y_n^s(b_j)/\lambda_j^n$. Then*

$$|Q_j(n) - c_j| \leq B_M(j) - c_j. \quad (6.1)$$

Proof. The leading spatial line is fixed by U_P , and its eigenvalue is $+\lambda_j$. Thus its contribution to $Q_j(n)$ is exactly c_j . For every other eigenvalue ξ of X_{b_j} , let P_ξ be its spectral projection. The unitary U_P commutes with that projection, so Cauchy–Schwarz gives

$$|\langle P_\xi v_{b_j}, U_P P_\xi v_{b_j} \rangle| \leq \|P_\xi v_{b_j}\|^2.$$

Consequently the absolute residual is at most

$$\sum_{\xi \neq \lambda_j} (|\xi|/\lambda_j)^n \|P_\xi v_{b_j}\|^2 \leq B_M(j) - c_j.$$

The series converges absolutely, being bounded by $\|v_{b_j}\|^2$. The kernel of X_{b_j} contributes zero because all exponents are positive. Negative spatial eigenvalues and odd powers are therefore included in the estimate. $\square$

One concentrated even moment therefore controls the normalized same-field pairing $Q_j(n)$ at every longer length, including odd lengths. To compare two temperatures, we use the common even reference length $M_j = k_{j+1}$. The lower bound for $r_j^{M_j}$ will cover the interval $M_j \leq n \leq 2M_j$.

Proposition 6.2. *For $j \geq 5$ and every integer n with $M_j \leq n \leq 2M_j$,*

$$\frac{Y_n^s(b_{j+1})}{Y_n^s(b_j)^2} \geq 1 - 4x_j. \quad (6.2)$$

Proof. Put $M = M_j$. At temperature b_j , Lemma 6.1 gives $Q_j(n) \leq B_M(j)$. This upper bound can be squared because the physical partition function, and hence $Q_j(n)$, is positive. At temperature b_{j+1} , use the same even length $M = k_{j+1}$:

$$Q_{j+1}(n) \geq 2c_{j+1} - B_M(j+1) \geq (1 - 2x_{j+1})B_M(j+1) > 0.$$

The last step uses Proposition 5.3. It follows that

$$\begin{aligned} \frac{Y_n^s(b_{j+1})}{Y_n^s(b_j)^2} &\geq (1 - 2x_{j+1})A(M, j)r_j^{n-M} \\ &\geq (1 - 2x_{j+1})(1 - x_j)(1 - u_j)(1 - u_{j+1}). \end{aligned} \quad (6.3)$$

Indeed $0 < r_j \leq 1$ and $0 \leq n - M \leq M$ give $r_j^{n-M} \geq r_j^M \geq (1 - u_j)(1 - u_{j+1})$. Using $x_{j+1} = 10x_j^2$, $u_j \leq x_j$, and $u_{j+1} \leq 6x_j^2$, the defect of the last product is at most

$$20x_j^2 + x_j + x_j + 6x_j^2 = 2x_j + 26x_j^2 \leq 4x_j,$$

as $x_j \leq .01$. This proves (6.2). $\square$

Proof of Theorem 1.1. Fix an odd n in $[M_j, 2M_j]$ with $j \geq 5$. Let $p_0 \geq p_1 \geq \dots$ be the Gibbs probabilities of H_n^s at inverse temperature b_j , listing energies with multiplicity. The left side of (6.2) is $\sum_i p_i^2$. Since $\sum_i p_i^2 \leq p_0 \sum_i p_i = p_0$, we have $p_0 \geq 1 - 4x_j > 1/2$. In particular the physical ground state is simple, and

$$e^{-b_j(E_1 - E_0)} = \frac{p_1}{p_0} \leq \frac{1 - p_0}{p_0} \leq \frac{4x_j}{1 - 4x_j} < 10^{-2^{j-5}}. \quad (6.4)$$

For the strict last inequality, use (5.12) and $4/[10(1 - .04)] < 1$. Because $b_j = 392 \cdot 2^{j-5}$, taking logarithms yields $E_1 - E_0 > \log(10)/392$.

Finally $M_5 = 1920$ and $M_{j+1} = 2M_j$. The intervals $[M_j, 2M_j]$, $j \geq 5$, cover every integer $n \geq 1920$. Every odd length $n = 2L + 1$ with $L \geq 960$ is therefore covered. Relabeling the sites gives precisely (1.1), with the single field $h = 3/5$ fixed in (2.1). $\square$

7 The boundary-selected infinite-volume state

The gap just proved supplies the hypothesis of Tasaki's topological-index theorem. We formulate the consequence for subsequential local limits, so that no convergence or uniqueness of the infinite-volume state is presupposed.

A local observable is a matrix acting on finitely many sites of $\mathbb{Z}$, extended by the identity on the other sites. A state on these observables is a normalized positive linear functional. Write ϕ_L for a normalized ground vector of $H_L(3/5)$ and set $\omega_L(A) = \langle \phi_L, A\phi_L \rangle$ whenever the support of A lies in $[-L, L]$. For $L \geq 960$ the vector is unique up to phase by Theorem 1.1, so this functional is well defined. Compactness of the density matrices on each finite interval, followed by a diagonal subsequence over the intervals, gives integers $L_k \rightarrow \infty$ for which

$$\omega(A) = \lim_{k \rightarrow \infty} \omega_{L_k}(A) \quad (7.1)$$

exists for every local observable. The limits are consistent on nested intervals and therefore define a state. We call any such state *boundary-selected*.

The bulk Hamiltonian is the formal sum $H_{\text{bulk}} = \sum_{j \in \mathbb{Z}} S_j \cdot S_{j+1}$. For a local A , its commutator $[H_{\text{bulk}}, A]$ is the finite sum of the commutators with bonds meeting the support of A . To specify the index, for $\varepsilon > 0$ define the local twist

$$U_\varepsilon = \exp \left[-i \sum_{j \in \mathbb{Z} \cap [-\pi/\varepsilon, \pi/\varepsilon]} (\pi + \varepsilon j) S_j^z \right]. \quad (7.2)$$

The rotations by 0 and 2π outside this interval act trivially for spin one. For the symmetric gapped states considered below, Tasaki's theorem guarantees the existence of the index $\text{Ind}(\omega) = \lim_{\varepsilon \downarrow 0} \omega(U_\varepsilon) \in \{-1, 1\}$; we use precisely the twist convention in [20, Equations (10)–(13)].

Corollary 7.1. *Every boundary-selected state (7.1) is invariant under uniform rotations about the z axis and under reflection $j \mapsto -j$. With $\gamma_* = (\log 10)/392$, it satisfies*

$$\omega(A^*[H_{\text{bulk}}, A]) \geq \gamma_*(\omega(A^*A) - |\omega(A)|^2) \quad \text{for every local } A. \quad (7.3)$$

In particular it is a locally-unique gapped ground state in the sense of [20, Definition 11], and its Tasaki index is -1 .

Proof. Both finite-volume symmetries commute with $H_L(3/5)$. Simplicity of the ground eigenspace implies invariance of ω_L under them; passing to (7.1) proves the asserted invariances.

For $L \geq 960$, the finite spectral theorem and Theorem 1.1 give

$$\langle A\phi_L, (H_L(3/5) - E_0(L, 3/5))A\phi_L \rangle \geq \gamma_*(\omega_L(A^*A) - |\omega_L(A)|^2).$$

The expression on the left is $\omega_L(A^*[H_L(3/5), A])$. Once the support of A and its adjacent sites lie strictly inside $[-L, L]$, this commutator equals $[H_{\text{bulk}}, A]$: the endpoint fields and all omitted bonds commute with A . Taking the limit proves (7.3). Its right side is nonnegative by the Cauchy–Schwarz inequality for a state; for $\omega(A) = 0$ it is the local gap inequality. The strict finite-volume lower bound has consequently yielded the non-strict lower bound γ_* in the limit.

Finally the Hamiltonians, spin normalization and endpoint-field signs are exactly those of [20, Equation (2)]. Theorem 1.1 verifies that paper’s Assumption 2 with the single field $h = 3/5$, $L_0 = 960$ and $\gamma = \gamma_*$. Its Theorem 3, with the subsequential interpretation specified in its End Matter, Equation (27), therefore applies to every state (7.1). To make the sign and limit order explicit, Tasaki’s finite-chain twist interpolation begins at the uniform π rotation. Its expectation is -1 , because the finite ground state has total $S^z = 1$ by that paper’s Lemma 1. Reflection makes the twist expectation real, and the uniform gap prevents it from crossing zero while the twist parameter is sufficiently small. More precisely, [20, Lemmas 6–8 and Equations (17)–(18)] gives, for every sufficiently small fixed $\varepsilon > 0$ and all sufficiently large L ,

$$-1 \leq \omega_L(U_\varepsilon) \leq -\sqrt{1 - \frac{4(\pi + \varepsilon)\varepsilon}{\gamma_*}}.$$

Here $L \geq \pi/\varepsilon$ ensures that the finite-chain twist equals the local operator (7.2). Passing first to (7.1) and then letting $\varepsilon \downarrow 0$ gives $\text{Ind}(\omega) = -1$. $\square$

The corollary evaluates the twist index for the selected states. It does not require the full sequence ω_L to converge or identify these states with every ground state of the infinite chain. In particular, the additional global uniqueness hypothesis of the half-chain conclusion in [20, Corollary 4] is not part of our result.

A Certified short-chain thermal moments

This appendix proves the two short-chain estimates used to initialize the boundary argument. Throughout, $h = 3/5$, $a = 700741/500000$, and $b_0 = 49/4$. All terminating decimals below denote exact rational numbers.

Proposition A.1. *Let λ_0 be the positive leading eigenvalue of X_{b_0} , and let c_0 be the squared norm of the projection of v_{b_0} onto its eigenspace. For*

$$P_*(x) = x^3 - .14x^2 - .18x,$$

one has

$$Y_6(b_0) < .027, \quad c_0 P_*(\lambda_0)^2 < .003745. \tag{A.1}$$

The first bound controls the residual contribution to $Y_m(b_0)$ in Lemma 5.2; the second controls the leading contribution. We first reduce both bounds to fifteen short-chain partition functions, then enclose these traces using matrices of order at most $3^6 = 729$.

A.1 Reduction to fifteen thermal traces

For $2 \leq n \leq 6$ and $g \in \{1, -1, 0\}$, define

$$\begin{aligned} H_{n,g} &= \sum_{j=0}^{n-2} \mathbf{S}_j \cdot \mathbf{S}_{j+1} - \frac{3}{5} S_0^z + \frac{3}{5} R_g, & R_g &= \begin{cases} g S_{n-1}^z, & g = 1, -1, \\ S_{n-1}^x, & g = 0, \end{cases} \\ Z_{n,g} &= \text{Tr} \exp[-b_0(H_{n,g} + a n I)]. \end{aligned} \quad (\text{A.2})$$

Thus $Z_{n,1} = Y_n(b_0)$ and $Z_{n,-1} = Y_n^s(b_0)$. The first bound in Proposition A.1 concerns $Z_{6,1}$. For the second, average the right-field direction over the rotation group.

Let $\mathcal{G}$ be the rotation group generated by the diagonal half-turns and the cyclic permutation of the coordinate axes, and let

$$\Pi = \frac{1}{|\mathcal{G}|} \sum_{\gamma \in \mathcal{G}} U_\gamma, \quad I(n) = \langle v_{b_0}, X_{b_0}^n \Pi v_{b_0} \rangle.$$

The group has order 12, and the orbit of z is the six signed coordinate axes, each with the same multiplicity. The boundary transfer identity therefore writes $I(n)$ as the average of the six corresponding right-field partition functions. Global rotations about z identify the four perpendicular orientations. Hence

$$I(n) = \frac{Z_{n,1} + Z_{n,-1} + 4Z_{n,0}}{6}. \quad (\text{A.3})$$

The averaging operator Π is an orthogonal projection commuting with X_{b_0} . By Proposition 3.4, $\mathcal{G}$ fixes the leading line pointwise. Therefore

$$\begin{aligned} c_0 P_*(\lambda_0)^2 &\leq \|P_*(X_{b_0}) \Pi v_{b_0}\|^2 \\ &= \langle v_{b_0}, P_*(X_{b_0})^2 \Pi v_{b_0} \rangle \\ &= I(6) - .28I(5) - .3404I(4) + .0504I(3) + .0324I(2). \end{aligned} \quad (\text{A.4})$$

In particular, no positivity of the individual odd powers of X_{b_0} is being assumed.

It therefore suffices to enclose $Z_{n,g}$ for $2 \leq n \leq 6$ and $g \in \{1, -1, 0\}$. The remaining subsections provide these enclosures and evaluate (A.4).

A.2 An integer matrix for the thermal approximation

Set

$$C_n = 3 \left\lfloor \frac{n-1}{2} \right\rfloor + 2((n-1) \bmod 2) + \frac{6}{5}, \quad A = I - \frac{H_{n,g} + C_n I}{8}. \quad (\text{A.5})$$

The operator A is a self-adjoint contraction. Here is a direct verification with the spin normalization in the theorem. A single bond has eigenvalues $-2, -1, 1$. For two consecutive bonds, couple the two outside spins to spin $t \in \{0, 1, 2\}$ and then couple this with the middle spin to total spin j . The operator

$$\mathbf{S}_1 \cdot \mathbf{S}_2 + \mathbf{S}_2 \cdot \mathbf{S}_3 = \mathbf{S}_2 \cdot (\mathbf{S}_1 + \mathbf{S}_3)$$

has eigenvalues $[j(j+1) - t(t+1) - 2]/2$, with $|t-1| \leq j \leq t+1$. Its minimum is -3 . Pairing consecutive bonds, with one single bond left over if necessary, and using the total boundary-field norm $6/5$ gives

$$0 \leq H_{n,g} + C_n I \leq \left(n - 1 + \frac{6}{5} + C_n \right) I \leq 16I.$$

Adding the lower bounds for the paired bonds is valid even where the three-site supports overlap. In order $n = 2, \dots, 6$, the displayed upper bounds are 5.4, 7.4, 10.4, 12.4, 15.4. Thus $-I \leq A \leq I$; positivity of A itself is not required.

Index the spin-weight basis by words $w = (w_0, \dots, w_{n-1})$ in $\{-1, 0, 1\}^n$, and let

$$p_w = \left(\sum_{j=0}^{n-1} w_j \right) \bmod 2 \in \{0, 1\}, \quad J_{ww} = \sqrt{1 + p_w}, \quad D = 80JAJ^{-1}.$$

The matrix D is integral. Indeed, its diagonal is

$$\eta(w) = 80 - 10C_n - 10 \sum_{j=0}^{n-2} w_j w_{j+1} + 6w_0 - 6gw_{n-1}. \quad (\text{A.6})$$

Each allowed bond hop $w \leftrightarrow w + \sigma(\mathbf{e}_j - \mathbf{e}_{j+1})$, $\sigma = \pm 1$, has entry -10 : the corresponding entry of the spin-one bond is 1, and such a hop preserves p_w . When $g = 0$, an allowed end hop from column $r = w + \sigma \mathbf{e}_{n-1}$ to row w has entry $-3(1 + p_w)$. To see this coefficient, the end-field entry before similarity is $3\sqrt{2}/10$, and the two parities are opposite, so

$$-10 \frac{3\sqrt{2}}{10} \sqrt{\frac{1 + p_w}{1 + p_r}} = -3(1 + p_w).$$

These are all the entries of D .

A.3 The exact recurrence

The recurrence approximates the half-temperature exponential

$$E = \exp[-b_0(H_{n,g} + C_n I)/2] = e^{-49} \exp(49A).$$

For a finite matrix M , write $\|M\|_{\text{F}}^2 = \text{Tr}(M^* M)$ for its squared Frobenius norm. Since E is self-adjoint, $\|E\|_{\text{F}}^2$ is the thermal trace $\text{Tr} \exp[-b_0(H_{n,g} + C_n I)]$. We approximate E by a normalized Poisson polynomial in the contraction A ; the similarity $D = 80JAJ^{-1}$ permits an integer Horner recurrence.

Put

$$K = 192, \quad Q = 2^{50}, \quad s_i = \frac{49^i K!}{i!}, \quad S_K = \sum_{i=0}^K s_i, \quad t_i = \left\lfloor \frac{Q s_i}{S_K} \right\rfloor. \quad (\text{A.7})$$

Starting with the zero matrix $\mathcal{X}_{K+1}$, compute

$$\mathcal{X}_i = \left\lfloor \frac{D \mathcal{X}_{i+1}}{80} \right\rfloor + t_i I, \quad i = K, K-1, \dots, 0. \quad (\text{A.8})$$

Matrix floors are entrywise, and every floor, including one with a negative argument, is toward $-\infty$. Define

$$\hat{E} = J^{-1} \frac{\mathcal{X}_0}{Q} J, \quad W_{n,g} = \|\hat{E}\|_{\text{F}}^2. \quad (\text{A.9})$$

For completeness, the following scalar version specifies the recurrence without matrix construction. Number words, if desired, by $\sum_j (w_j + 1) 3^j$, and interpret a lookup outside $\{-1, 0, 1\}^n$ as

zero. Initialize $x(w, r) = 0$ for all pairs of words. At stage $i = K, \dots, 0$, use the current array to form

$$\begin{aligned}
z(w, r) &= \eta(w)x(w, r) - 10 \sum_{j=0}^{n-2} \sum_{\sigma=\pm 1} x(w + \sigma(\mathbf{e}_j - \mathbf{e}_{j+1}), r) \\
&\quad - \mathbf{1}_{\{g=0\}} 3(1 + p_w) \sum_{\sigma=\pm 1} x(w + \sigma \mathbf{e}_{n-1}, r), \\
x_{\text{new}}(w, r) &= t_i \mathbf{1}_{\{w=r\}} + \left\lfloor \frac{z(w, r)}{80} \right\rfloor.
\end{aligned} \tag{A.10}$$

Replace all entries simultaneously. At the end, the required table entry is the integer quotient

$$k_{n,g} := \left\lfloor 10^{16} W_{n,g} \right\rfloor = \left\lfloor \frac{10^{16} \sum_{w,r} x(w, r)^2 (2 - p_w)(1 + p_r)}{2Q^2} \right\rfloor. \tag{A.11}$$

The weight follows from $(1 + p_r)/(1 + p_w) = (2 - p_w)(1 + p_r)/2$. All operations in (A.7), (A.10), and (A.11) are integer operations; they may be performed with arbitrary-precision integers.

The resulting fifteen entries are

$n \backslash g$	1	-1	0
2	10185296075298	4160017987	583578389575
3	259254948723	571111660440989	47796720049308
4	318278534893	661347985	43971049724
5	260908126900	29917222640283	5659378873682
6	16467340956	328219854	4086161507

(A.12)

Equations (A.7), (A.10), and (A.11) provide a finite exact-arithmetic verification of every entry.

A.4 Error of the matrix approximation

Lemma A.2. *For every pair (n, g) in (A.12),*

$$\left\| \widehat{E} - \exp[-b_0(H_{n,g} + C_n I)/2] \right\|_{\text{F}} < 2 \cdot 10^{-10}.$$

Proof. Write $d = 3^n$. Since $b_0/2 = 49/8$, the exact target is

$$E = e^{-49} \exp(49A) = \sum_{i=0}^{\infty} e^{-49} \frac{49^i}{i!} A^i.$$

The polynomial with coefficients $q_i = s_i/S_K$ is the truncation of this Poisson expansion, renormalized to have coefficient sum one. Its coefficient ℓ^1 error is twice the omitted Poisson mass, at most $2R$, where

$$R = \frac{49^{193}}{193!} \frac{1}{1 - 49/194} < 3.152 \cdot 10^{-33}. \tag{A.13}$$

In bounding the omitted Poisson mass by R , we have discarded the factor e^{-49} . This coefficient estimate is used in [6, proof of Proposition 1]; here it is applied to powers of a contraction. Because A is a contraction, the corresponding Frobenius error is at most $2\sqrt{d} R$.

At each Horner stage the coefficient replacement $q_i \mapsto t_i/Q$ contributes less than $\sqrt{d}/Q$ in Frobenius norm. The entrywise floor contributes less than d/Q before undoing the similarity, and hence less than $\sqrt{2}d/Q$ afterwards, since $\|J^{-1}\| \|J\| \leq \sqrt{2}$. Subsequent propagation is left multiplication by A and cannot increase the Frobenius norm. Thus the contraction is used after undoing the similarity; no contraction property of the generally nonsymmetric integer matrix D is asserted. The total error is less than

$$2\sqrt{d}R + \frac{193(\sqrt{d} + \sqrt{2}d)}{Q} \leq 54R + \frac{193(27 + \sqrt{2}729)}{2^{50}} < 54R + \frac{432513}{2^{51}} < 2 \cdot 10^{-10}. \quad (\text{A.14})$$

The penultimate inequality uses only $\sqrt{2} < 3/2$; the last one is a rational comparison using (A.13). $\square$

Squaring the Frobenius norm converts this estimate into a thermal trace. In fact, E is self-adjoint and $\|E\|_F^2 = \text{Tr} \exp[-b_0(H_{n,g} + C_n I)]$. Consequently

$$\left| Z_{n,g} - e^{b_0(C_n - an)} W_{n,g} \right| < e^{b_0(C_n - an)} \left(4 \cdot 10^{-10} \sqrt{W_{n,g}} + 4 \cdot 10^{-20} \right). \quad (\text{A.15})$$

This follows from $|\|E\|_F^2 - \|\hat{E}\|_F^2| \leq 2\|E - \hat{E}\|_F \|\hat{E}\|_F + \|E - \hat{E}\|_F^2$.

A.5 Rational intervals from the printed table

The displayed integer floors suffice to verify the needed estimates; no unprinted digits of $W_{n,g}$ are used. For an entry $k = k_{n,g}$, put

$$L = \frac{k}{10^{16}}, \quad U = \frac{k+1}{10^{16}}, \quad r = \frac{\lfloor \sqrt{k+1} \rfloor + 1}{10^8}, \quad \varepsilon = 4 \cdot 10^{-10}r + 4 \cdot 10^{-20}. \quad (\text{A.16})$$

Then $L \leq W_{n,g} < U$ and $\sqrt{W_{n,g}} < r$. In every entry $L - \varepsilon > 0$, so (A.15) gives

$$e^{b_0(C_n - an)}(L - \varepsilon) < Z_{n,g} < e^{b_0(C_n - an)}(U + \varepsilon). \quad (\text{A.17})$$

The square root in (A.16) is just an integer square root. To enclose the remaining exponential by rational numbers, set

$$T_{80}(y) = \sum_{j=0}^{80} \frac{y^j}{j!}.$$

For $0 \leq y \leq 10$, comparison of consecutive tail terms gives

$$T_{80}(y) \leq e^y \leq T_{80}(y) + \frac{2y^{81}}{81!}. \quad (\text{A.18})$$

For a negative argument, take reciprocals and reverse the endpoints. All arguments $b_0(C_n - an)$ here lie between $-.0544635$ and 9.691073 . Thus (A.16)–(A.18) require only rational arithmetic and integer square roots. In particular they give

$$.0266319440 < Z_{6,1} < .0266319607 < .027. \quad (\text{A.19})$$

Using (A.17) and (A.18) in (A.3) gives the following rational centers, each with absolute error less than 10^{-7} :

n	2	3	4	5	6
center of $I(n)$	.02703074	.01203562	.01193511	.00931560	.00893272.

(A.20)

Substitution of these centers in the right-hand side of (A.4) gives .003744031780. The sum of the absolute polynomial coefficients is 1.7032, so the total possible error is less than $1.7032 \cdot 10^{-7}$. The right-hand side of (A.4) is consequently less than

$$.003744031780 + .00000017032 = .0037442021 < .003745.$$

Together with (A.19), this proves Proposition A.1.

B Exact trial vectors for opposite fields

We prove the four variational inequalities used to initialize the boundary argument. Throughout this appendix,

$$a = \frac{700741}{500000}, \quad h = \frac{3}{5}, \quad \alpha = \frac{2021}{5000} = .4042,$$

and the unshifted opposite-field Hamiltonian on m sites is

$$H_m^{\text{opp}} = \sum_{j=1}^{m-1} \mathbf{S}_j \cdot \mathbf{S}_{j+1} - \frac{3}{5} S_1^z + \frac{3}{5} S_m^z. \quad (\text{B.1})$$

The bulk matrices below are those of the periodic variational construction in [16, Section “Two variational inputs”]. We give their complete integer specification and the new endpoint vectors, so the present finite estimates can be reproduced without importing a variational bound for open chains.

Proposition B.1. *Let $E_0^{\text{opp}}(m)$ be the smallest eigenvalue of H_m^{opp} . Then*

$$E_0^{\text{opp}}(m) + am < \alpha \quad \text{for } m \in \{34, 78, 142, 240\}. \quad (\text{B.2})$$

The proof consists of a fixed integer matrix-product vector, an exact contraction for its norm and energy, and four integer comparisons. No spectral approximation enters the calculation.

B.1 The complete tensor and endpoint data

Let $\mathcal{V}$ have the 26 labels

$$(i, d), \quad 1 \leq i \leq 8, \quad d = -\ell_i, -\ell_i + 2, \dots, \ell_i, \quad (\ell_i)_{i=1}^8 = (1, 1, 1, 1, 3, 3, 3, 5). \quad (\text{B.3})$$

Order them first by i and then by increasing d . Three real 26-by-26 matrices A_s , $s \in \{-1, 0, 1\}$, are specified as follows. At row (i, d) and column (j, e) put $\ell = \ell_i$, $k = \ell_j$, and $r = (\ell + d)/2$. The entry is zero unless

$$e = d + 2s, \quad |k - \ell| \leq 2, \quad k = \ell \implies i = j. \quad (\text{B.4})$$

For an allowed entry, multiply the coefficient in the second column of Table 1 by the factor in its s column. The parameters D_i and P_{ij} appear in Tables 2 and 3. The latter parameter is defined only when $\ell_j = \ell_i + 2$. All labels in these rules must belong to (B.3); an out-of-range column is absent.

Define columns $\mathbf{b}, \mathbf{c} \in \mathbb{R}^{26}$ by

$$\mathbf{b}_{(i,d)} = \begin{cases} \beta_i, & d = -1, \\ 0, & d \neq -1, \end{cases} \quad \mathbf{c}_{(i,d)} = \begin{cases} \kappa_i \beta_i, & d = -1, \\ 0, & d \neq -1. \end{cases} \quad (\text{B.5})$$

All their parameters are included in Table 2.

For each $m \geq 2$, the physical vector is

$$\psi_m = \sum_{s_1, \dots, s_m = -1}^1 (\mathbf{b}^\dagger A_{s_1} \cdots A_{s_m} \mathbf{c}) e_{s_1} \otimes \cdots \otimes e_{s_m}. \quad (\text{B.6})$$

This is the open-boundary matrix-product form of [17, Section 3.1, Equation (11)], with endpoint rows and columns absorbed into the first and last matrices. The vectors e_{-1}, e_0, e_1 are the orthonormal spin basis fixed in the statement of the problem. In particular, the coefficients in (B.6) are ordinary real amplitudes: no virtual inner product or physical basis weights are inserted. Normalization of the matrices A_s is unnecessary. The support rule forces a nonzero amplitude to have $\sum_j s_j = 0$, because both endpoints have virtual weight -1 . This restriction on the trial vector does not restrict the variational upper bound on the smallest eigenvalue of the full Hamiltonian.

The following tables complete the integer specification of these matrices and endpoints.

	coefficient	$s = -1$	$s = 0$	$s = 1$
$k = \ell + 2$	P_{ij}	$2(\ell + 1 - r)(\ell + 2 - r)$	$2(r + 1)(\ell - r + 1)$	$(r + 1)(r + 2)$
$k = \ell$	D_i	$2(\ell - r + 1)$	d	$-(r + 1)$
$k = \ell - 2$	$-P_{ji}$	2	-2	1

Table 1: Every allowed entry of A_s is the coefficient times the listed factor. All other entries are zero.

i	ℓ_i	D_i	β_i	κ_i
1	1	-289	10000	12
2	1	-84	5620	12
3	1	-9	2456	12
4	1	33	4868	12
5	3	-22	82	6
6	3	27	-809	6
7	3	71	12475	6
8	5	-11	-9645	1

Table 2: Virtual types, diagonal tensor parameters, and endpoint parameters. The left and right endpoints are supported on $d = -1$.

(i, j)	P_{ij}	(i, j)	P_{ij}	(i, j)	P_{ij}
(1, 5)	-11	(2, 7)	-49	(4, 6)	9
(1, 6)	-35	(3, 5)	-6	(4, 7)	-32
(1, 7)	-28	(3, 6)	-25	(5, 8)	1
(2, 5)	-23	(3, 7)	-20	(6, 8)	2
(2, 6)	16	(4, 5)	-34	(7, 8)	-16

Table 3: All fifteen couplings with $\ell_j = \ell_i + 2$.

B.2 Norm, fields, and the bond insertion

For $s, t \in \{-1, 0, 1\}$, put $B_{st} = A_s A_t$ and define

$$T_* = \sum_{s=-1}^1 A_s \otimes A_s, \quad G_* = \sum_{s=-1}^1 s A_s \otimes A_s, \quad (\text{B.7})$$

$$\begin{aligned} O_* &= \sum_{s,t=-1}^1 st B_{st} \otimes B_{st} \\ &+ \sum_{\substack{s=-1,0 \\ t=0,1}} (B_{st} \otimes B_{s+1,t-1} + B_{s+1,t-1} \otimes B_{st}). \end{aligned} \quad (\text{B.8})$$

We use the usual Kronecker convention $(C \otimes D)_{(v,w),(v',w')} = C_{v,v'} D_{w,w'}$. Write

$$L_0 = (\mathbf{b} \otimes \mathbf{b})^t, \quad u = \mathbf{c} \otimes \mathbf{c}, \quad L_i = L_0 T_*^i. \quad (\text{B.9})$$

Lemma B.2. *With the data above, the norm $V_m = \langle \psi_m, \psi_m \rangle$ and the scaled shifted energy numerator*

$$H_m^* = 500000 \langle \psi_m, (H_m^{\text{opp}} + amI) \psi_m \rangle$$

satisfy

$$V_m = L_m u, \quad (\text{B.10})$$

$$\begin{aligned} H_m^* &= 500000 \sum_{r=0}^{m-2} L_0 T_*^r O_* T_*^{m-2-r} u - 300000 L_0 G_* T_*^{m-1} u \\ &+ 300000 L_{m-1} G_* u + 700741 m L_m u. \end{aligned} \quad (\text{B.11})$$

In particular, V_m and H_m^ are integers.*

Proof. Expanding T_*^m sums the products of two identical real amplitudes over all physical words. This proves (B.10) directly in the orthonormal physical basis. Inserting G_* at one site instead of T_* multiplies the summand by the weight s and therefore inserts S^z at that site.

For a bond, the spin normalization gives

$$\mathbf{S} \cdot \mathbf{S} = S^z \otimes S^z + \frac{1}{2}(S^+ \otimes S^- + S^- \otimes S^+).$$

Its diagonal entry on (s, t) is st . Its off-diagonal entries are 1 between (s, t) and $(s+1, t-1)$ for $s \in \{-1, 0\}$ and $t \in \{0, 1\}$, since each allowed ladder entry is $\sqrt{2}$. The two summands in each parenthesis of (B.8) give the two directions of that matrix element. Thus O_* inserts exactly one unit-coupling bond, with no extra factor or metric. Its position after r sites gives the r th summand of (B.11).

The left and right endpoint insertions have coefficients $500000(-3/5) = -300000$ and $500000(3/5) = 300000$, respectively. The constant shift contributes $500000am = 700741m$ times the norm. There are $m-1$ bond positions and no closing bond, giving the stated formula. Every matrix, row, and column in that formula is integral. $\square$

B.3 The invariant 162-dimensional contraction

All contractions in Lemma B.2 can be carried out in a single 162-dimensional space. On a paired virtual label $((i, d), (j, e))$, consider the difference $d - e$. Each nonzero entry of A_s changes virtual weight by $2s$ from row to column. Consequently every summand $A_s \otimes A_s$ of T_* or G_* preserves this difference. A nonzero entry of B_{st} changes weight by $2(s + t)$. The two factors in every tensor summand of O_* have the same total physical weight, including the pair (s, t) and $(s + 1, t - 1)$ in the off-diagonal terms. Each such summand also preserves $d - e$.

Let $\mathcal{K}_0$ be the coordinate subspace with paired labels

$$\mathcal{Q} = \{((i, d), (j, d)) : (i, d), (j, d) \in \mathcal{V}\}. \quad (\text{B.12})$$

The multiplicities of the virtual weights $d = -5, -3, -1, 1, 3, 5$ are 1, 4, 8, 8, 4, 1, respectively; hence

$$\dim \mathcal{K}_0 = 1^2 + 4^2 + 8^2 + 8^2 + 4^2 + 1^2 = 162. \quad (\text{B.13})$$

The preceding argument shows that T_*, G_*, O_* are block diagonal by weight difference. Both L_0 and u are supported in $\mathcal{K}_0$, because the two endpoint vectors are supported on $d = -1$. We may therefore restrict these three matrices to (B.12) before multiplying them, taking powers, or forming the expressions of Lemma B.2. In particular, squaring the restricted T_* gives the restriction of T_*^2 ; there is no contribution through a discarded state. From now on the same symbols denote these restricted integer matrices and endpoint tensors, ordered by the inherited lexicographic order.

B.4 An exact row recurrence

The energy sum in (B.11) need not be evaluated separately at every bond. Define integer rows by

$$\begin{aligned} L_i &= L_{i-1}T_* \quad (i \geq 1), \\ N_1 &= -300000L_0G_*, \\ N_i &= N_{i-1}T_* + L_{i-2}(500000O_* + 700741T_*^2) \quad (i \geq 2). \end{aligned} \quad (\text{B.14})$$

Then

$$V_m = L_mu, \quad H_m^* = (N_m + 300000L_{m-1}G_* + 700741L_m)u. \quad (\text{B.15})$$

To verify the latter identity, induction on m in (B.14) gives

$$N_m = -300000L_0G_*T_*^{m-1} + \sum_{r=0}^{m-2} L_0T_*^r(500000O_* + 700741T_*^2)T_*^{m-2-r}. \quad (\text{B.16})$$

Its bond contributions and left field are exactly those in (B.11). Each of its $m - 1$ shift contributions equals $700741L_m$. The final term $700741L_m$ in (B.15) supplies the remaining site shift, and the other final term supplies the right field with the required positive sign. This proves (B.15).

For specificity, the whole contraction can be implemented by keeping just three rows. Begin with

$$x = L_0, \quad y = L_0T_*, \quad w = -300000L_0G_*.$$

For each $m = 2, 3, \dots, 240$, make the following updates, whose right-hand sides all use the old rows:

$$\begin{aligned} w_{\text{new}} &= wT_* + 500000xO_* + 700741(xT_*)T_*, \\ x_{\text{new}} &= y, \quad y_{\text{new}} = yT_*. \end{aligned} \quad (\text{B.17})$$

After this iteration, $(x, y, w) = (L_{m-1}, L_m, N_m)$. At a required length compute

$$V = yu, \quad H = wu + 300000x(G_*u) + 700741V. \quad (\text{B.18})$$

Thus $(V, H) = (V_m, H_m^*)$. The initialization realizes the same invariant at $m = 1$, which proves the indexing of the algorithm. Ordinary exact integer matrix multiplication suffices throughout; sparsity can reduce the work but changes no operation or comparison.

B.5 The integer comparisons

Table 4 gives the results of (B.17)–(B.18). The integers p_m , q_m , and k_m in the table have the following exact meanings:

$$\begin{aligned} 10^{p_m-1} &\leq V_m < 10^{p_m}, \\ q_m 10^{p_m-6} &\leq V_m < (q_m + 1) 10^{p_m-6}, \\ k_m V_m &\leq 20H_m^* < (k_m + 1)V_m. \end{aligned} \quad (\text{B.19})$$

The last comparison is equivalent to $k_m = \lfloor 10^7 H_m^* / (500000V_m) \rfloor$, since $10^7 / 500000 = 20$. The first comparison in particular certifies that the vector ψ_m is nonzero.

m	p_m	q_m	k_m	$H_m^* / (500000V_m)$ lies in
34	202	715867	4031974	[.4031974, .4031975)
78	440	747265	4041175	[.4041175, .4041176)
142	787	121288	4041110	[.4041110, .4041111)
240	1317	165352	4041006	[.4041006, .4041007)

Table 4: Exact integer contraction certificates. The intervals are rational intervals obtained from integer division, with strict upper endpoints.

All four strict comparisons

$$202100V_m - H_m^* > 0 \quad (m = 34, 78, 142, 240) \quad (\text{B.20})$$

follow as well. They are recorded directly in the full integer output, and they also follow from the last line of (B.19), since every displayed $k_m + 1$ is less than $4042000 = 20 \cdot 202100$.

The accompanying computational files make the integer comparison fully reproducible. The program `verification/trial_verify.py` constructs the matrices from the data in this appendix and executes the row recurrence with arbitrary-precision integers. The file `verification/trial-exact-matrices.json` contains every entry of A_s , the endpoint vectors, the restricted paired labels, and the matrices T_* , G_* , O_* . The file `verification/trial-exact-results.json` contains the full integers V_m , H_m^* , and $202100V_m - H_m^*$ for each row, rather than only their shortened display in Table 4. The equations and tables of this appendix specify the calculation independently of these files.

Proof of Proposition B.1. For each of the four lengths, the integer $V_m = \langle \psi_m, \psi_m \rangle$ is positive by (B.19). The variational principle, Lemma B.2, and (B.20) give

$$E_0^{\text{opp}}(m) + am \leq \frac{\langle \psi_m, (H_m^{\text{opp}} + amI)\psi_m \rangle}{\langle \psi_m, \psi_m \rangle} = \frac{H_m^*}{500000V_m} < \frac{202100}{500000} = \alpha.$$

This proves all four inequalities with the same fixed field $h = 3/5$. $\square$

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